

DEPARTMENT OF CSE**PROJECT ABSTRACT SUBMISSION****COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
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TITLE OF THE PROJECT	DeployIQ – Intelligent CI/CD Pipeline Optimizer	
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Abstract

DeployIQ is an intelligent CI/CD Pipeline Optimizer designed to improve the speed, reliability, and cost-efficiency of software delivery pipelines through data-driven analysis and machine learning. Traditional CI/CD pipelines are often built once and left largely unmonitored, resulting in redundant build steps, poor resource utilization, unnecessarily long build and test cycles, and late detection of failures. DeployIQ addresses these problems by continuously collecting and analyzing CI/CD logs and historical build data from GitHub Actions workflows to identify bottlenecks, recommend concrete optimizations, and predict builds that are likely to fail before they consume further compute time.

The system supports multiple user roles, including Developers, DevOps Engineers, and Administrators. Developers can view pipeline health and failure-risk scores for their commits and pull requests, DevOps Engineers can inspect stage-level performance metrics, apply optimization recommendations such as test parallelization and build caching, and configure GitHub Actions workflow integrations, while Administrators manage users, repository onboarding, and organization-wide pipeline policies. A machine learning model trained on historical build logs learns recurring failure patterns and flags high-risk builds ahead of time, and a centralized dashboard visualizes build trends, resource consumption, and failure predictions across repositories.

The project follows the **Agile Scrum Software Development Life Cycle (SDLC)**, allowing iterative development through sprint planning and continuous improvement. Development activities will be managed using **Jira** and **GitHub**, while **Docker** and **Kubernetes** will be used for containerization and deployment. Code quality and security will be ensured through **SonarQube** and **OWASP ZAP**, and the application will be deployed on a cloud platform such as AWS or Google Cloud.

The proposed system aims to provide a secure, scalable, and intelligent solution for optimizing GitHub Actions-based CI/CD pipelines, reducing build times and infrastructure costs while improving delivery reliability for development and DevOps teams through machine learning-driven failure prediction.